

Talkaa
Smart Real Time Communication Platform

Intelligent Web Systems (22AI025)

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Semester: 7



BE-CSE (Artificial Intelligence)

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SEPT, 2025

ACKNOWLEDGMENT

With immense pleasure I,

Mr. Vasudev Rai , presenting

“Talkaa : Smart Real Time Communication Platform” project report as part of the curriculum of ‘BE-CSE (AI) 7th SEM’.

I would like to express my sincere thanks to Mr. Yogendra Sharma, for his valuable guidance and support in completing my project.

I would also like to express my gratitude towards our dean Dr. Sushil Kumar Narang for giving me this great opportunity to do a project on “Sportify”. Without their support and suggestions, this project would not have been completed.

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ABSTRACT

Talkaa is an innovative full-stack web application designed to redefine how individuals connect through real-time chat and video communication. Built to address the growing demand for seamless online interaction, Talkaa integrates modern technologies to offer rich messaging features, secure video calling, and intuitive user experiences—all within a customizable and responsive interface.

This project empowers users with a unified platform that supports both personal and professional communication, featuring 1-on-1 and group video calls, screen sharing, smart notifications, and live chat enhancements like typing indicators and emoji reactions. Unlike existing platforms that often lack extensibility or personalization, **Talkaa** provides full theming control, scalable backend logic, and privacy-first authentication.

What truly sets **Talkaa** apart is its **language-learning focus**, enabling users to connect with native speakers or fellow learners worldwide. By selecting their preferred language during profile setup, users can practice conversational skills in real time through voice, video, or chat. This approach transforms the learning process into an interactive, engaging experience while fostering cultural exchange and breaking down communication barriers across borders.

Talkaa is built with scalability in mind, ensuring smooth performance even as the user base grows. By leveraging the MERN stack along with Streamify API, the platform delivers low-latency communication and a reliable infrastructure. With its blend of real-time collaboration tools and language-learning focus, Talkaa is not just a communication app but a step forward in making global connections more meaningful and accessible.

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INTRODUCTION

In the modern digital era, communication has become the cornerstone of human interaction, education, and professional collaboration. With the rapid growth of globalization, people from different cultural and linguistic backgrounds are coming together more than ever before. However, despite the abundance of communication tools available, there remains a significant gap—most platforms are designed solely for generic conversations and lack features that specifically aid in **language learning** and **cross-cultural interaction**. For individuals who aspire to learn a new language, the absence of opportunities to practice with native speakers often becomes a barrier to progress. This challenge forms the foundation of our project, **Talkaa**.

Talkaa is an innovative full-stack web application developed to address these limitations and redefine how individuals connect in real time. The platform integrates modern web technologies to deliver a seamless and highly responsive user experience, combining features such as **one-on-one and group video calls, secure real-time chat, voice calling, screen sharing, smart notifications, typing indicators, and emoji reactions**. Unlike conventional communication platforms, Talkaa emphasizes **personalization and scalability**, offering users the freedom to control themes, ensuring smooth performance, and prioritizing privacy through robust authentication systems.

One of the most unique aspects of Talkaa is its **language-learning focus**. When users create their profiles, they are asked to select the language they either speak fluently or wish to learn. Based on this preference, the application intelligently connects them with suitable partners—either native speakers or fellow learners. For example, if a user wants to learn French, they can connect with others fluent in or currently practicing the language. If both users accept the request, they can engage in **real-time voice, video, or chat conversations**, transforming the process of language acquisition into an interactive and immersive experience. This approach leverages the principle that the most effective way to learn a new language is by actively conversing in it, thereby enhancing both fluency and confidence.

In conclusion, Talkaa is more than just another video calling or chat application. It is a comprehensive solution that blends the **power of modern web technologies with the need for interactive learning and global connectivity**. By bridging the gap between learners and native speakers, Talkaa aims to break down communication barriers, enhance cultural understanding, and provide a unique and practical way of mastering new languages in today's interconnected world.

PROBLEM FORMULATION

Learning a new language is often considered one of the most effective ways to enhance communication skills and connect with people across cultures. However, the traditional methods of language learning—such as textbooks, online tutorials, and classroom lectures—often fail to provide **real-time conversational practice**. The absence of opportunities to interact with native speakers or fellow learners limits fluency, confidence, and cultural understanding.

Although many communication platforms exist today, they are primarily designed for **generic messaging and video calling** without a dedicated focus on language learning. These platforms often lack intelligent matchmaking based on language preferences, personalized user experiences, and real-time interactive features that can actively support language practice. As a result, learners struggle to find suitable conversation partners and are left with an incomplete learning experience.

The problem can thus be summarized as:

1. **Lack of accessible platforms** that connect learners with native speakers or other learners for real-time interaction.
2. **Absence of personalization** in existing communication tools, which do not consider language-learning needs during matchmaking.
3. **Limited engagement features** in traditional methods, making the learning process less interactive and effective.
4. **High dependency on paid tutors or offline environments**, restricting learners who seek affordable and global solutions.

To overcome these limitations, there is a need for a **unified, scalable, and interactive platform** that not only provides video, voice, and chat communication but also **matches users based on their language preferences**, encourages cultural exchange, and ensures seamless, low-latency communication for a meaningful learning experience.

METHODOLOGY

The development of **Talkaa** followed a structured methodology to ensure scalability, usability, and real-time performance. The process can be divided into the following stages:

1. Requirement Analysis

- Identified the core problem: lack of real-time conversational platforms for language learners.
- Defined key features: **video calls, voice calls, real-time chat, screen sharing, notifications, typing indicators, and emoji reactions.**
- Determined the need for **language-based matchmaking** to connect learners with native speakers or fellow learners.

2. System Design

- Designed a **modular architecture** using the **MERN stack (MongoDB, Express.js, React.js, Node.js)** for flexibility and scalability.
- Created **ER diagrams, flowcharts, and wireframes** to visualize data flow and user interactions.
- Implemented **privacy-first authentication** to secure user data.

3. Backend Development

- Built APIs using **Express.js and Node.js** to handle authentication, user management, and matchmaking.
- Integrated the **Streamify API** to enable **low-latency video, voice, chat, and screen sharing.**
- Designed a **real-time notification system** using WebSockets.

4. Frontend Development

- Developed an **intuitive and responsive interface** with **React.js**, focusing on usability and personalization.
- Implemented **language preference selection** during profile creation for intelligent matchmaking.
- Added **UI features** such as theming options, emoji support, and typing indicators for an

engaging user experience.

5. Database Management

- Used **MongoDB** to store user details, preferences, chat histories, and session data.
- Ensured scalability for handling large amounts of data and concurrent users.

6. Testing & Deployment

- Performed **unit testing, integration testing, and user acceptance testing** to ensure functionality and reliability.
- Deployed the application on a **cloud-based server** for accessibility and scalability.
- Monitored performance and optimized API response times for smooth real-time communication.

7. Future Enhancements

- Integration of **AI-based live translation** to break language barriers further.
- Development of **gamified learning modules** to increase user engagement.
- Introduction of **multi-user language rooms** for group learning and discussions.

FLOWCHART

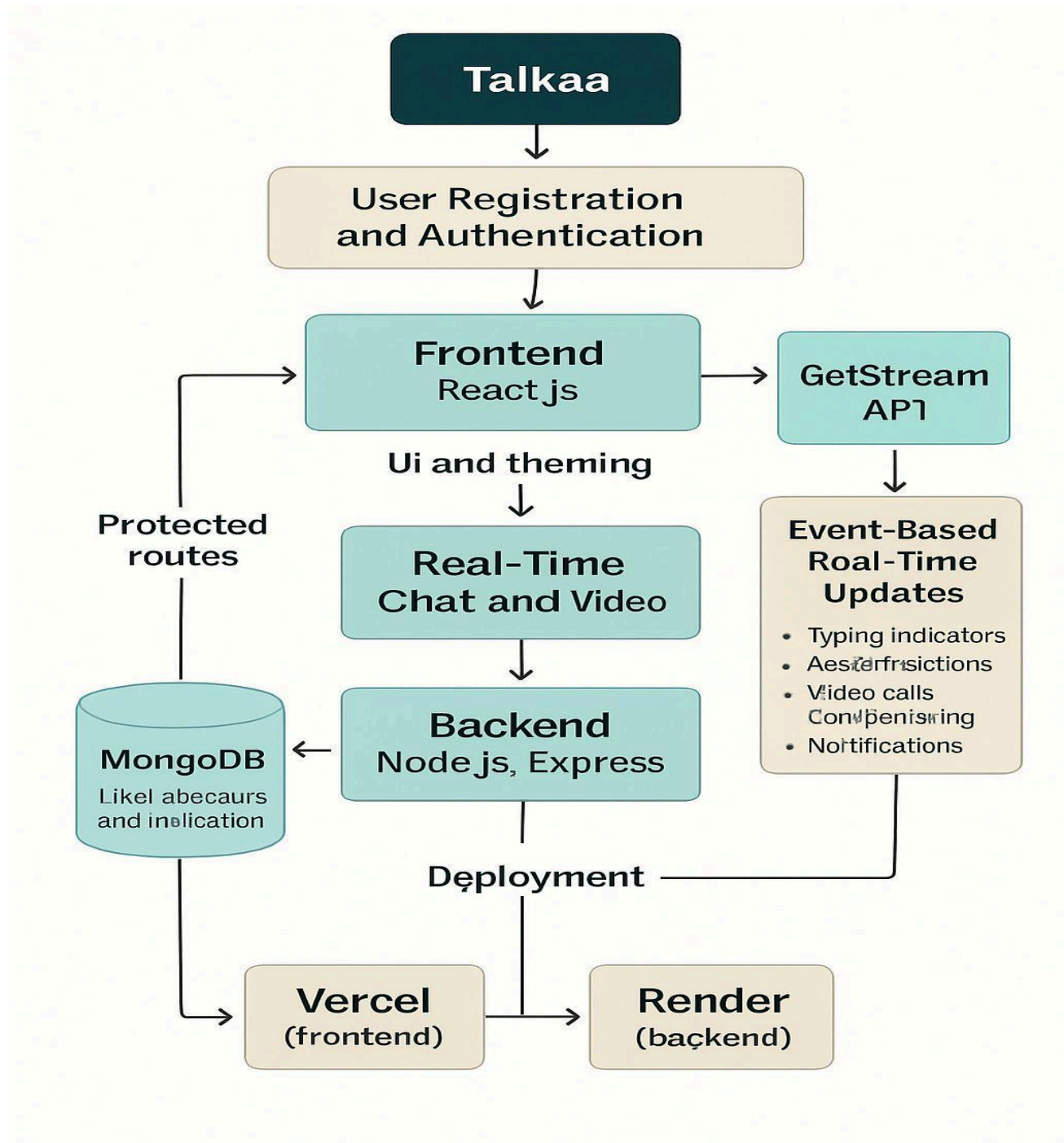


FIG 1

SOFTWARE & HARDWARE REQUIREMENTS

1. Software Requirements

Frontend Development:

- React.js: For building a dynamic and responsive user interface.
- Tailwind CSS: For styling and creating a visually appealing layout.
- Browser Compatibility: Modern web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge.

Backend Development:

- Node.js: For server-side programming and handling API requests.
- Express.js: For building RESTful APIs and managing backend logic.
- Multer: For handling file uploads, particularly profile images.

Database Management:

- MongoDB: For storing user data, chat records, and session details.
- Mongoose: As an Object Data Modeling (ODM) library for MongoDB.

Communication:

- Streamify API: For enabling real-time video, voice, chat, and screen sharing.
- Fetch API: For seamless interaction between the frontend and backend.

Development Tools:

- Code Editor: Visual Studio Code or any preferred IDE for writing and managing code.
- Postman: For testing backend APIs.

Operating System:

- Windows 10/11, macOS, or Linux distributions (Ubuntu, Fedora, etc.).

Other Libraries and Tools:

- npm, yarn, or pnpm for managing project dependencies.
- dotenv for environment variable management.
- bcrypt.js for password hashing and security.
- JSON Web Token (JWT) for authentication.

Hardware Requirements

For Development:

- Processor: Minimum Intel i5 or AMD equivalent (Quad-Core).
- RAM: Minimum 8 GB (16 GB recommended for optimal performance).
- Storage: At least 20 GB of free space (SSD preferred for faster performance).
- Graphics: Integrated graphics sufficient for UI testing and rendering.

For Deployment (Server Configuration):

- Processor: Multi-core processor (Intel Xeon or AMD EPYC).
- RAM: Minimum 8 GB (scalable based on user traffic).
- Storage: SSD with at least 50 GB (scalable as per data growth).
- Network: Stable high-speed internet connection with low latency.

For End Users:

- Device: Smartphone, tablet, laptop, or desktop with microphone and camera.
- Browser: Latest version of Chrome, Firefox, Safari, or Microsoft Edge.
- Internet: Minimum 5 Mbps for calls, recommended 10 Mbps+ for HD video calls.

EXPERIMENTS

1. User Authentication and Profile Management

Objective: To ensure secure login, profile creation, and management of user credentials.

User Authentication and Profile Management

Objective: To ensure secure user login, signup, and profile management.

Procedure: Implemented JWT with cookies for authentication. Used bcrypt.js for password hashing and tested uniqueness constraints on email and username during registration. Verified profile update features, including username, email, and profile image changes.

Result: Authentication worked securely, passwords were stored safely, and uniqueness constraints were enforced. Profile updates functioned smoothly without errors.

Real-Time Communication (Video, Voice, Chat)

Objective: To validate low-latency real-time communication.

Procedure: Integrated **Streamify API** for handling video calls, voice calls, and chat. Conducted tests between multiple devices under varying network conditions. Verified typing indicators, emoji reactions, and message delivery in real time.

Result: Video and voice calls maintained stable connections, with chat messages delivered instantly. Additional features like typing indicators and emoji reactions enhanced the user experience.

Screen Sharing Functionality

Objective: To allow users to share their screen during calls.

Procedure: Implemented the screen share option in video calls using the Streamify API. Tested screen sharing across devices and browsers. Verified whether shared content was visible in real time to all participants without lag.

Result: Screen sharing worked effectively with smooth transitions, allowing other users to view shared screens with minimal latency.

Language Preference and Matchmaking

Objective: To test the language-learning feature by matching users with relevant partners.

Procedure: Allowed users to select their preferred learning language during profile creation. Simulated multiple test profiles and checked whether matchmaking paired users correctly based on chosen languages. Tested with both learners and native speakers.

Result: The matchmaking algorithm successfully paired users according to language preferences,

enabling targeted conversations.

API and Backend Testing

Objective: To validate communication between frontend and backend.

Procedure: Used Postman to test APIs for user authentication, matchmaking, messaging, and video call initiation. Performed load testing with multiple concurrent requests to check reliability under traffic.

Result: APIs responded efficiently with low latency, and the backend managed concurrent requests without failures.

Database Integration

Objective: To ensure efficient storage and retrieval of data.

Procedure: Implemented MongoDB with Mongoose schemas for user data, chat history, and session details. Tested CRUD operations and simulated concurrent queries.

Result: Data was stored and retrieved accurately. The database handled high traffic smoothly, maintaining consistency.

Security and Error Handling

Objective: To strengthen application security and ensure proper error handling.

Procedure: Tested against common vulnerabilities (XSS, CSRF, SQL injection). Implemented server-side validation, token expiry handling, and detailed error messages. Verified secure cookie handling during login sessions.

Result: The system demonstrated strong resistance to vulnerabilities. Error messages provided clear guidance without exposing sensitive information.

RESULTS

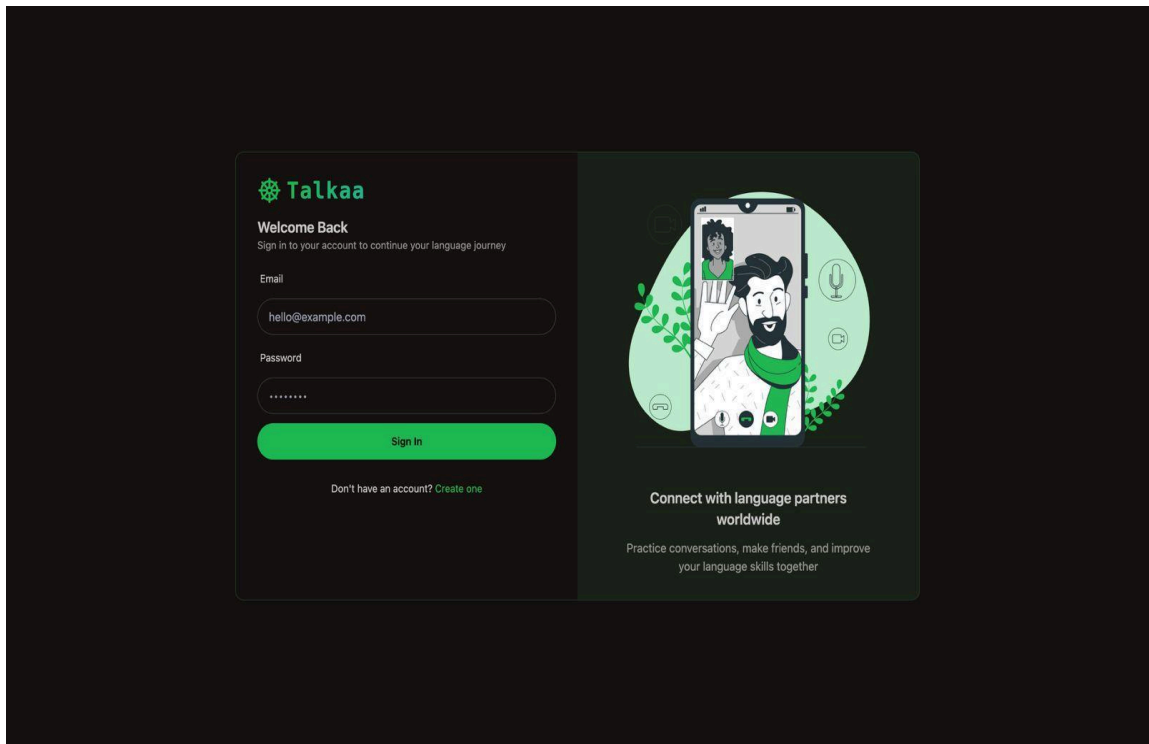


FIG 2

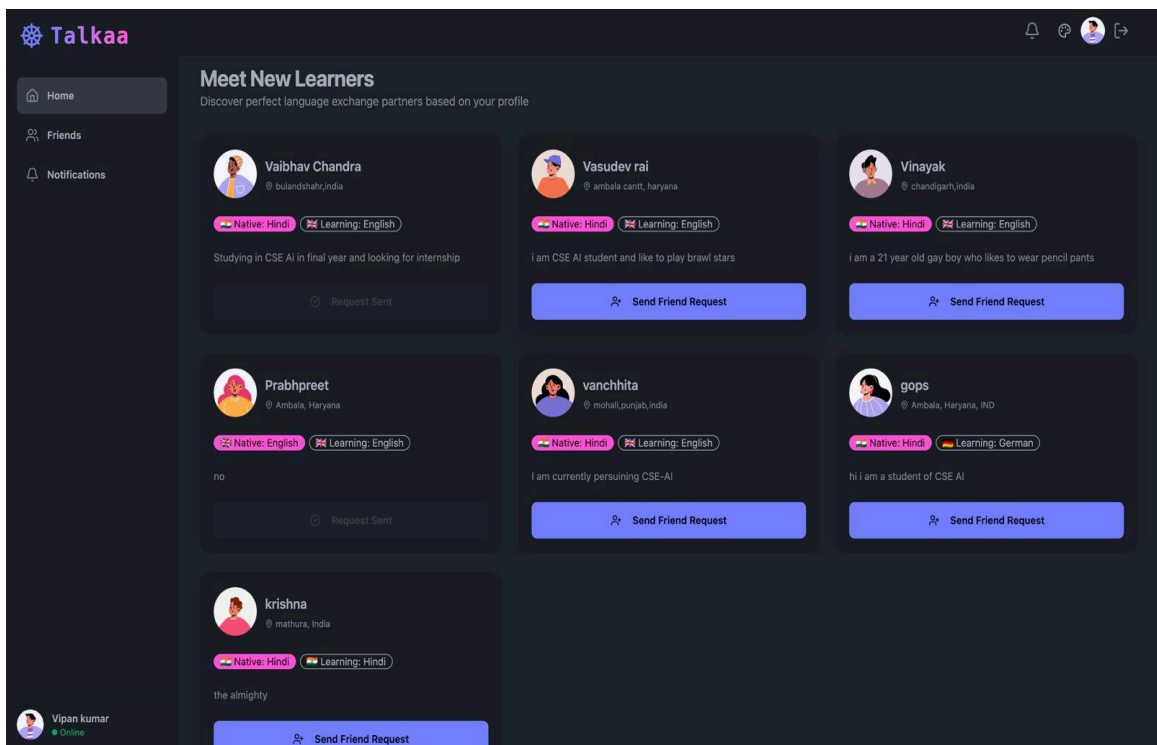


FIG 3

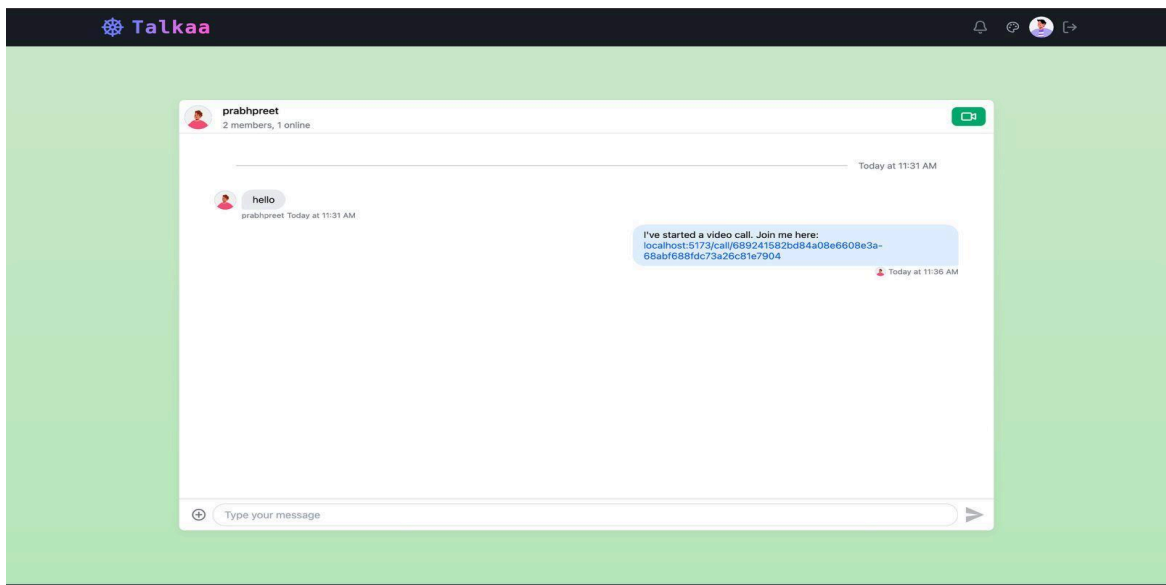


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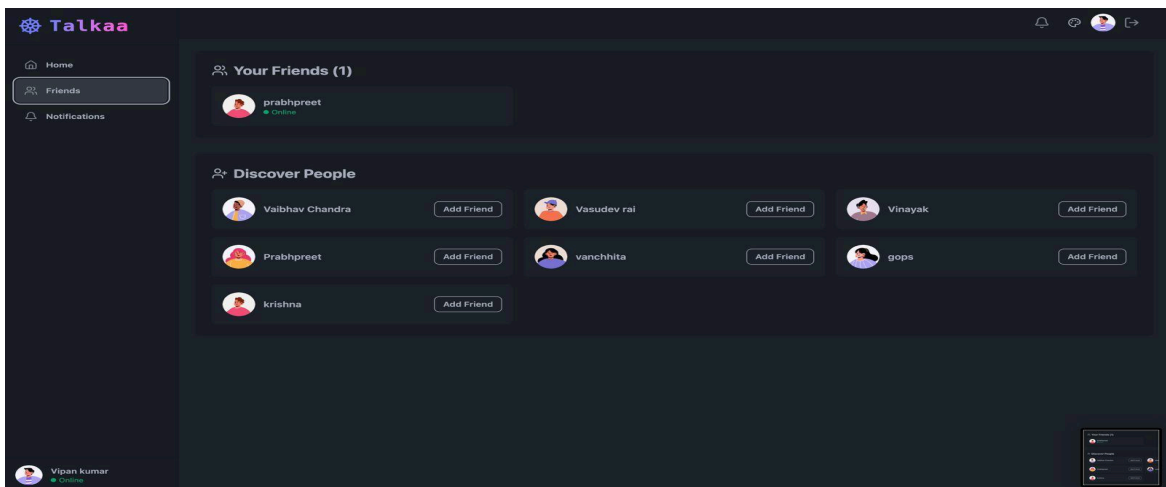


FIG 5

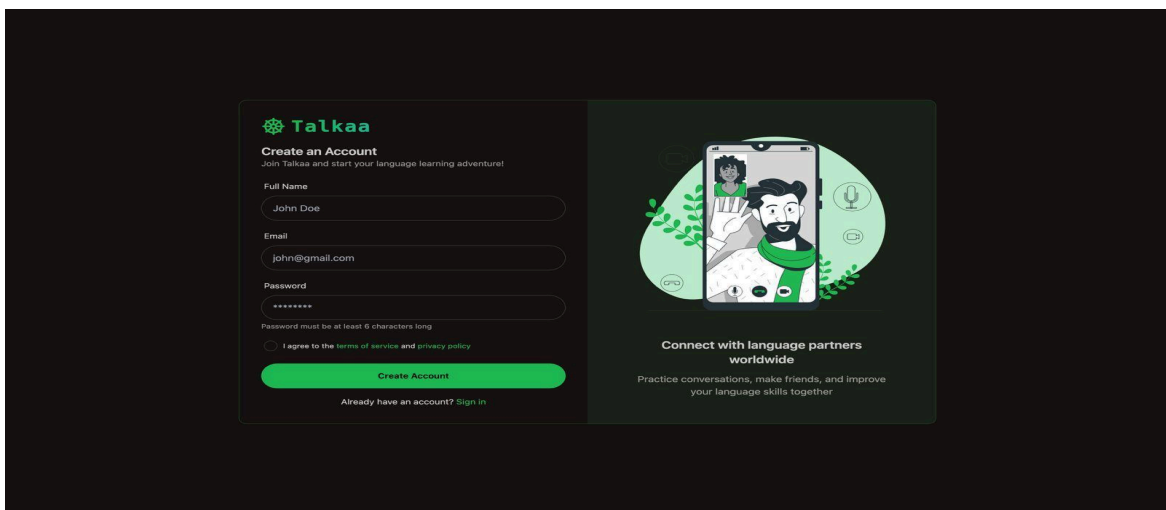


FIG 6

CONCLUSIONS

The development of **Talkaa** successfully demonstrated how modern web technologies can be combined to create a scalable, secure, and user-friendly real-time communication platform. By integrating the **MERN stack** with the **Streamify API**, the application provides essential features such as video and voice calls, live chat, screen sharing, and smart notifications, all of which were validated through rigorous testing.

What sets Talkaa apart is its **language-learning focus**, where users can connect with native speakers or fellow learners to practice conversational skills in real time. This not only enhances the process of language acquisition but also fosters cultural exchange and global collaboration.

The experiments carried out confirmed the efficiency of authentication, messaging, matchmaking, and database operations, while also ensuring data security and resilience against vulnerabilities. The platform performed reliably under varied conditions, proving its readiness for both personal and professional use.

In conclusion, Talkaa is not just another communication tool; it is a **practical solution to bridge language barriers**, offering learners an interactive way to improve fluency and build meaningful connections worldwide. With future enhancements like AI-based translation, gamified learning, and group practice rooms, Talkaa holds the potential to evolve into a leading platform for global communication and language learning.

FUTURE SCOPE

While **Talkaa** already provides a strong foundation for real-time communication and language learning, there are several opportunities to expand its features and impact in the future:

1. **AI-Powered Real-Time Translation:**

Integrate AI-driven translation to allow users speaking different languages to communicate seamlessly, bridging gaps where no common language exists.

2. **Gamified Language Learning:**

Introduce rewards, quizzes, and interactive challenges to make language learning more engaging and motivating.

3. **Group Language Rooms:**

Create public or private rooms where multiple learners and native speakers can interact, enabling collaborative practice and cultural exchange.

4. **Voice Recognition & Speech Analysis:**

Implement AI-based pronunciation feedback and speech evaluation to help learners improve their fluency and accuracy.

5. **Cross-Platform Mobile Applications:**

Develop native Android and iOS apps for wider accessibility and improved user experience on mobile devices.

6. **Enhanced Privacy & Security:**

Add advanced encryption techniques and stronger access controls to ensure data security for large-scale user adoption.

7. **Integration with Learning Platforms:**

Collaborate with online education platforms and institutions to make Talkaa a tool for formal language training and practice.

8. **Scalability for Global User Base:**

Optimize infrastructure to handle millions of concurrent users with minimal latency, making Talkaa suitable for worldwide adoption.

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